

Xiang (Shawn) Fei

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EDUCATION

Carnegie Mellon University

Ph.D. in Mechanical Engineering | Advisor: Prof. Sebastian Scherer

Pittsburgh, PA

Aug. 2026 - Aug. 2031

Carnegie Mellon University

M.S. in Robotics | Advisor: Prof. Howie Choset

Pittsburgh, PA

Aug. 2024 - May. 2026

The Chinese University of Hong Kong (Shenzhen)

B.Eng. in Computer Science and Engineering

Shenzhen, China

Sept. 2020 - May 2024

University of California, Berkeley

Visiting Student

Berkeley, CA

Jan. 2023 - May 2023

INTERESTS, SKILLS

Interests of Research: Robotics; Simultaneous Localization and Mapping (SLAM); Deep Learning; Multi-sensor Fusion

Skills: ROS, Python, C++, C, Matlab, Pytorch, TensorFlow, CUDA, Isaac Sim, Gazebo, PyBullet, Raspberry Pi, Autodesk Fusion, Unreal Engine, Blender, Linux, Git, Latex

PUBLICATIONS

* Equal contribution

X. Fei*, Y. Qiu*, C. Xu, Y. Chen, R. Li, W. Wang, S. Scherer. *MAC-I²: Learned Metrics-Aware Covariance for Robust Visual-Inertial Initialization and Calibration*. (Under Review).

Y. Yang*, **X. Fei***, L. Wang*, C. Li, Z. Dai, H. Kou, L. Li, H. Choset. *DeformX: A Versatile Co-Simulation Framework for Deformable Linear Objects*. 2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).

X. Fei, T. Tian, H. Choset, L. Li. *Bag of Word Groups (BoWG): A Robust and Efficient Loop Closure Detection Method Under Perceptual Aliasing*. 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). [[Paper](#), [Code](#), [Website](#)]

X. Fei, H. Zhao, X. Zhou, J. Zhao, T. Shu, F. Wen. *Power System Fault Diagnosis with Quantum Computing and Efficient Gate Decomposition*. Scientific Reports. [[Paper](#)]

Y. Cao, X. Zhou, **X. Fei**, H. Zhao, W. Liu, J. Zhao. *Linear-Layer-Enhanced Quantum Long Short-Term Memory for Carbon Price Forecasting*. Quantum Machine Intelligence. [[Paper](#), [Code](#)]

X. Zhou, H. Zhao, Y. Cao, **X. Fei**, G. Liang, J. Zhao. *Carbon Market Risk Estimation Using Quantum Conditional Generative Adversarial Network and Amplitude Estimation*. Energy Conversion and Economics. [[Paper](#)]

X. Zhou, X. Wang, **X. Fei**, W. Liu, B. Xie, J. Zhao. *Carbon Disclosure Effect, Corporate Fundamentals, and Net-zero Emission Target: Evidence from China*. Under Review. [[Paper](#)]

Data Analysis Team: **X. Fei**, H. Bi, S. Dai, Y. Xu, C. Tao. Carbon Rating Report of China's 100 Overseas Listed Companies. (REPORT) 2022 Global Forum on Sustainable Development. [[Report](#)]

RESEARCH EXPERIENCE

AirLab, Robotics Institute, Carnegie Mellon University

Graduate Research Assistant. Advised by Prof. Sebastian Scherer

Pittsburgh, PA

Oct. 2024 - Present

- Research Topics: SLAM with multi-sensor fusion and learning-based methods.

Biorobotics Lab, Robotics Institute, Carnegie Mellon University

Graduate Research Assistant. Advised by Prof. Howie Choset

Pittsburgh, PA

Jun. 2023 - Present

- Research Topics: SLAM in confined environments using laser-based structured light; Loop closure under perceptual aliasing; Deformable linear objects simulation, perception and manipulation; Human-robot motion retargeting.

Energy Internet Lab, The Chinese University of Hong Kong (Shenzhen)

Research Assistant. Advised by Prof. Junhua Zhao

Shenzhen, China

Feb. 2024 - May. 2024

- Research Topics: Power system; Smart grid; Quantum machine learning and optimization.

Winning Health Technology Group Co., Ltd.

Algorithm Engineer Intern (Computer Vision Track)

Shanghai, China

Jul. 2022 - Sept. 2022

- Developed learning-based 3D reconstruction of gastrointestinal tract using monocular endoscope.

ACTIVITIES

- Teaching Assistant, The Chinese University of Hong Kong (Shenzhen)** 2022
- Served as Teaching Assistant for CSC3001 Discrete Mathematics.
- Peer Advisor, The Chinese University of Hong Kong (Shenzhen)** 2022
- Mentored junior students on academic planning and campus life; recipient of the Outstanding Peer Advisor Award.

HONORS & AWARDS

- **China National College Students Competition on Energy Economics**
 - 2023 National Outstanding Award of Graduate Group (National Top 10) May 2023
 - 2022 National Outstanding Award of Undergraduate Group (National Top 5) May 2022
- **CCF “Sinan Cup” Quantum Computing Programming Competition**
 - 2023 Third Prize at Finals Jul. 2023
 - 2022 Second Prize at Finals (International Top 18) Aug. 2022
- **2022 Mathematical Contest in Modeling**
 - Meritorious Winner (Global Top 7%) May 2022
- **2020-2023 Dean’s List Honor of School of Data Science, CUHKSZ** Sept. 2021 - 2023
- **2021-2022 Academic Performance Scholarship of CUHKSZ** Dec. 2022
- **Undergraduate Research Award of CUHKSZ** Nov. 2022
- **2020-2021 Master’s List Award of Muse College, CUHKSZ** Nov. 2021
- **Bowen Scholarship of CUHKSZ** Sept. 2020